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Novel Hydrogen Cell Factories for Sustainable Energy Generation and Wastewater Management

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Abstract

This study aims to evaluate the feasibility of using a renewable and cost-effective system called Microbial Electrolysis Cell (MEC) for produced water (PW) treatment while recovering energy in the form of hydrogen gas. The generated hydrogen can serve as an energy-neutral/positive source for waste treatment in the gas-oil-separation plant (GOSP). Additionally, the output water, with reduced oil content, can be repurposed for re-injection operations to support reservoir pressure and for fracturing operations. This study aims to identify key parameters that impact MECs performance for PW treatment. The assessment included a detailed analysis of the PW treatment process, focusing on (1) microbial community analysis of the electroactive biofilm and metabolic pathways; (2) testing different applied voltages while monitoring electroactive biofilm growth through chronoamperometry and cyclic voltammetry; and (3) conducting abiotic control and open circuit voltage (OCV) control experiments. A sampling protocol was developed to systematically collect inoculum from industrial environments. This protocol aimed to enrich microbial communities capable of degrading specific contaminants in PW, thereby enhancing their effectiveness in the microbial electrolysis process. MEC oxidize organic carbon contaminants at the anode, releasing electrons that are used to generate energy in the form of hydrogen gas at the cathode. The energy required to drive the reaction in the MEC is lower than the conventional voltage required for water splitting (i.e., 1.8–3.5 V). Hence, the process can be powered by renewable energy sources, such as wind or solar, which enhances the plant's water handling capacity. This study explores different testing strategies and reactor configurations to optimize PW treatment and hydrogen production from enriched microbial communities. which could be further optimized for larger-scale applications. In light of the industrial efforts towards energy mix and transition, MEC technology can generate green electricity with low CAPEX to design and construct the bioreactor in the field, and helps to address one of the pressing business and environmental challenges in the upstream operations by (1) Providing clean energy-neutral source from PW; (2) Generating non-damaging water for injection and fracturing operations; (3) Unlocking oil potential by increasing plant capability in handling water. It also includes an analysis of energy consumption and energy balance. The samples were processed for microbial communities profiling through DNA Nanopore

sequencing technology and bioinformatic analysis, revealing novel microbial species that are crucial for MEC performance. Additionally, analytical and electrochemical assessments helped in identifying suitable setups for producing electrical current in MECs fed with PW and Disposal Water (DW), as well as optimal methods for contaminants breakdown. The varying concentrations of PW and DW (0.5-3%) generated electrical current in MECs, indicating their potential as energy sources, demonstrating that both PW and DW contain sufficient organic matter to support bio-electrochemical processes, which could be further optimized for larger-scale applications. In light of the industrial efforts towards energy mix and transition, MEC technology can generate green electricity with low CAPEX to design and construct the bioreactor in the field, and helps to address one of the pressing business and environmental challenges in the upstream operations by (1) Providing clean energy- neutral source from PW; (2) Generating non-damaging water for injection and fracturing operations; (3) Unlocking oil potential by increasing plant capability in handling water.

Introduction

PW includes freshwater and/or seawater that has been trapped alongside oil and gas in geological reservoirs, as well as water that is injected into wells with the aim to increase hydrocarbon recovery rates. The geographical location, age, depth of geological formation, and hydrocarbon forming geochemistry play an important role in determining the physical and chemical properties of the PW. The high concentration of salt content, total dissolved solids (TDS), total suspended solids (TSS), chemical oxygen demand (COD), and alkalinity in PW makes it more difficult to treat compared to other types of wastewaters [1]. Additionally, PW also contains a mixture of organic and inorganic compounds, such as dissolved and dispersed hydrocarbons, TPH, BTEX, oil and grease, PAH, phenols, heavy metals, treatment chemicals, dissolved gases, waxes, nitrates, and sulfates [2]. Such compounds in PW must be removed before it can be safely discharged into the environment or reused for injection. It is estimated that the amount of PW associated with hydrocarbon recovery is more than 250 million barrels per day globally, making it the largest waste stream associated with oil and gas activity [3]. In addition, as global energy demands continue to rise, the amount of PW generated is expected to increase each year. In addition, the composition of PW can damage the permeability of near-wellbore formations when it is reused for injection, resulting in decreased well injectivity and economic losses for the industry [4]. Conventional methods for treating PW, such as physical, chemical, and membrane-based technologies are expensive, energy intensive and have several limitations. Physical and chemical treatments require high retention times, have high capital costs, and generate significant amounts of secondary waste whose disposal is also problematic. On the other hand, membrane-based technologies have limitations regarding the types of contaminants they can remove and are susceptible to fouling, which can result in increased maintenance and replacement costs. All the above-mentioned issues represent significant challenges for the oil and gas companies, which are looking to reduce their environmental impact and lower operating costs. It is worth noting that conventional treatment methods, in addition to being energy- intensive, do not recover the energy that is embedded in the PW, which can be a valuable source of renewable energy if it can be effectively captured and utilized. Therefore, there is a global need to develop more sustainable and energy-efficient methods of PW management and treatment not only to minimize its impact on the environment and reduce the operational costs. but also recovers and utilizes the energy it contains. Thus, this research study aims to explore an innovative and energy-efficient method for the treatment of PW with recovery of energy as hydrogen in MECs.

Microbial Electrolysis Cells (MEC)

The proposed solution for the treatment of PW involves using bioelectrochemical systems (BESs), specifically Microbial Electrolysis Cells (MECs), which can effectively treat PW while generating energy-rich hydrogen gas. This process can lead to significant energy savings and economic benefits for the

company. BES, such as Microbial Fuel Cells (MFCs) and MECs, are innovative bio-electrochemical technologies that utilize electroactive microorganisms to convert chemical energy stored in organic and inorganic compounds into electricity or hydrogen through catalytic oxidation reactions (Figure 1) [5]. BESs such as MFC have been proven effective in removing hydrocarbons, sulfates, BTEX, and other compounds present in PW [2, 6, 7]. However, our study proposes an innovative alternative that utilizes MECs not only for the treatment of PW but also for the recovery of energy embedded in the wastewater as hydrogen, which can be utilized as a valuable source of renewable energy. Compared to conventional wastewater treatment technologies, MEC is an anaerobic technology that does not require energy for aeration systems. Also, due to the low energy input needed to drive the bioelectrochemical process, these systems can be powered by renewable sources of energy, such as wind and solar, and have negligible impact on the environment. MECs can efficiently remove contaminants such as organic compounds, total dissolved solids, and sulfates, while also enabling the recovery of the energy embedded in the PW. A successful pilot-scale testing of MEC technology has been conducted treating real wastewater and high salinity water, including brine pools [8, 9, 5, 10]. During these tests, hydrogen gas and methane gas were successfully recovered, demonstrating the potential for treatment of high salinity wastewater and scalability of the technology while generating energy sources. The effluent water quality from PW treatment has a great influence on the injectivity of injection and disposal wells. Ideally, injected water should be compatible with the formation water and should not contain suspended solids or oil in order to prevent formation damage and thus maintain long term injectivity [4]. MECs are versatile and can be combined with other technologies, such as membrane technology [11], to further improve the quality of the treated water effluent. This would be beneficial for the purpose of re-injecting PW. In any electrochemical process, high electrolyte salinity is preferred to decrease resistance and increase conductivity, which is a crucial factor for scale-up and commercial applicability [12]. The high salinity of PW increases conductivity for current production, making MECs a viable technology for the treatment of PW and transfer of electrons for hydrogen gas production. Consequently, the high salinity of PW, which poses a challenge for conventional treatment technologies, can serve as an advantage for electrochemical methods. The production of hydrogen through waste valorization has become increasingly important due to its clean, renewable, and sustainable approach [13]. Therefore, the proposed solution will offer significant insights into the potential of MECs for the oil and gas industry for both, the treatment of produced water and the generation of renewable energy. This study aims to evaluate the feasibility of using MECs for the treatment of PW with inherent recovery of energy in the form of hydrogen gas, focusing on identifying the key design and operation parameters that will impact the performance of MECs for PW treatment. Ultimately, this study provides a cost-effective, sustainable, and environmentally friendly solution for the treatment of PW and the generation of renewable energy, in addition to providing practical recommendations for the scale up and commercial implementation of the technology in the oil and gas industry.

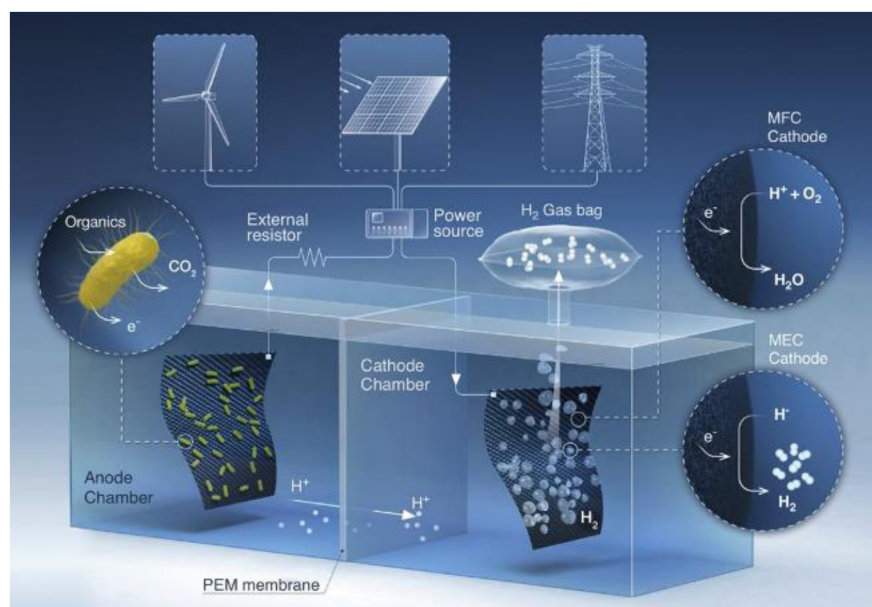


Figure 1—Schematic of the different configurations of bioelectrochemical systems for wastewater treatment with recovery of energy. MFC refers to microbial fuel cell; MEC refers to microbial electrolysis cell. PEM refers to proton exchange membrane. Given the low energy input to drive the bio electrochemical reaction, BESs can be powered by renewable sources of energy such as wind and solar [14].

This study aims to explore the use of MECs for the treatment of PW and the generation of energy-rich hydrogen gas, which has significant potential for the oil and gas industry. PW is a major environmental challenge for the industry, and globally, a substantial number of resources is spent treating trillions of liters of PW every year. By demonstrating the feasibility of using MECs for the treatment of PW and the generation of hydrogen gas, our study will further address this significant environmental challenge while delivering cost savings and environmental benefits as follows:

Cost savings: The global market for produced water treatment was valued at 9.10 Billion USD in 2021 and is projected to reach 14.86 Billion USD by 2030, growing at a compound annual growth rate of 5.2% from 2022 to 2030 [15]. Adopting energy-neutral or positive wastewater treatment systems such as MECs and reducing the need for expensive conventional treatment methods can result in significant cost savings for the oil and gas industry, which spends a substantial amount each year on PW treatment.

Generation of renewable energy: The generation of hydrogen gas from PW using MECs provides a new source of energy from waste for the oil and gas industry. By producing hydrogen from PW, the industry can reduce its reliance on fossil fuels, thereby mitigating its contribution to greenhouse gas emissions and supporting the transition to a low-carbon economy. Furthermore, hydrogen production using MECs presents a valuable opportunity for revenue generation and opens up the possibility of developing a tradable commodity. As the global market for hydrogen continues to grow, this could provide a medium- to long-term commercial advantage for the industry while contributing to the overall reduction of greenhouse gas emissions.

Environmental impact mitigation: Implementing water efficient practices, maximizing wastewater reuse, and minimizing water losses are part of the oil and gas industry efforts towards a comprehensive water conservation and management. The use of MECs can result in significant cost savings for the industry by reducing the volume of PW that needs to be disposed of, increase the reuse water and addressing the growing challenge of PW management.

Materials and Methods

Sampling of PW

In this study, produced water (PW), disposal water (DW) and soil samples were collected from three different oil and gas production sites, serving as inoculum sources and substrates for reactor operations. The sample containers were shipped under cold conditions and arrived at King Abdullah University of Science and Technology (KAUST). Upon arrival, the samples were immediately stored at 4°C to prevent changes in the quality of the samples. Careful selection of the initial microbial community or bacterial inoculum was crucial to ensure that it mirrors the characteristics of the contaminants in the PW that needs to be treated. In particular, using microorganisms sourced from PW is imperative due to their adaptation to extreme conditions such as high salt concentrations.

Assessment phase

During this phase, laboratory experiments were conducted by setting up MEC reactors and evaluating their performance in treating PW and producing hydrogen gas. Different applied potentials were tested in parallel while monitoring the growth of electroactive biofilms using techniques such as chronoamperometry and cyclic voltammetry. Other parameters have been monitored over the course of the experiments, including current density, energy/gas production (CH₄ and H₂), organics (COD), total petroleum hydrocarbons (TPH), benzene, toluene, ethyl benzene, and xylene (BTEX), sulfates (SO₄⁻), nitrogen (NH₄⁺, NO₂⁻, NO₃⁻), and total dissolved solids (TDS).

Biocatalysts characterization

Microbial community from oilfield samples have been analyzed for 16S rRNA and metabolic pathways responsible for removing the contaminants and producing energy through metagenomic analyses. Moreover, abiotic control and open circuit voltage (OCV) control experiments were conducted to evaluate the role of the microbial activity in the removal of contaminants and energy production.

DNA extraction and library preparation. DNA extraction from the PW, DW and soil samples was performed using a slightly modified protocol based on the FastDNA Spin Kit for Soil (MP Biomedicals, USA). Gel electrophoresis was performed using the TapeStation 2200 and Genomic DNA Screentapes (Agilent, USA) to validate the size and purity of a subset of DNA extracts. DNA concentrations were measured using the Qubit dsDNA HS/BR Assay Kit (Thermo Fisher Scientific, USA). Amplicon libraries targeting the 16S/18S rRNA gene variable regions 4-8 (abeV48-A) for archaea, bacteria, and eukaryota were prepared using a custom protocol. Up to 25 ng of extracted DNA was used as a template for PCR amplification. The forward and reverse primers used included custom 24 nt barcode sequences and the following target sequences for abeV48-A: [515FB] GTGYCAGCMGCCGCGGTAA and [1391R] GACGGGCGGTGWGTRCA (Apprill et al., 2015 & Parada et al., 2016).

Optimization phase

This process focused on improving the performance of the MECs and identifying the optimal operational conditions to enhance removal rates and hydrogen production. Various start-up strategies were tested to determine the best approach for acclimatizing the microbial community to the PW. Additionally, evaluated different reactor configurations (such as single and double chamber reactors) to determine their effectiveness in hydrogen recovery. Finally, energy consumption and energy balance analyses were conducted to gain a better understanding of the overall energy efficiency of the system.

Results and Discussion

DNA extraction and sequencing

DNA extraction from the two different types of PW did not yield measurable concentrations, suggesting either a very low microbial load or the presence of substances that inhibit DNA extraction, such as microbial inhibitors used during the oil extraction and treatment processes. In contrast, soil samples yielded sufficient DNA concentrations and were further processed for sequencing library preparation. The resulting amplicon libraries were purified using the Clean NGS SPRI beads (CleanNA, NL) protocol, with a bead-to-sample ratio of 3:5, and DNA was eluted in 25 µL of nuclease-free water (Qiagen, Germany). Sequencing libraries were prepared from the purified amplicons using the SQK-LSK114 kit (Oxford Nanopore Technologies, UK) following the manufacturer's protocol. DNA concentrations were measured using the Qubit dsDNA HS Assay kit (Thermo Fisher Scientific, USA), and gel electrophoresis was performed using the TapeStation 2200 and D1000/High Sensitivity D1000 screen tapes (Agilent, USA) to validate the size and purity of a subset of the amplicon libraries.

Bioinformatic processing

The sequencing reads in the demultiplexed and base called fastq files were filtered for length (320 - 2000 bp) and quality (phred score > 17) using a local implementation of filtlong v0.2.1 with the settings –min_length 320 –max_length 2000 –min_mean_q 98. The SILVA 16S/18S rRNA 138 SSURef NR99 full-length database in RESCRIPt format was downloaded from the QIIME on [16, 17, 18]. Potential generic place holders and dead-end taxonomic entries were cleared from the taxonomy flat file. The filtered reads were mapped to the SILVA 138.1 99 % NR database with minimap2 v2.24-r1122 using the -ax map-ont command [19] and downstream processing using samtools v1.14 [20]. Mapping results were filtered such that query sequence length relative to alignment length deviated < 5 %. Noteworthy, low-abundant OTUs making up < 0.1 % of the total mapped reads within each sample were disregarded as a data denoising step. Further bioinformatic processing was done via RStudio IDE (2023.12.1.402) running R version 4.4.1 (2024-06-14) and using the R packages: ampvis2 (2.8.7) [21], tidyverse (2.0.0), seqinr (4.2.36), ShortRead (1.60.0) and iNEXT (3.0.1) [22, 23].

The sequencing statistics (Table 1) provide key information, including DNA concentration and various metrics related to sequencing quality and diversity of the inoculum samples. Raw reads refer to the total number of DNA reads generated from sequencing, while FiltReads represent the number of reads remaining after filtering for quality and length. MappedReads indicate the final number of reads mapped to the reference database after filtering for mapping quality, and the is the proportion of MappedReads to FiltReads. OTUs (Operational Taxonomic Units) reflect the total number of unique OTUs observed in the sample, and Richness represents the total number of unique OTUs. The Shannon and Simpson indices are widely used alpha-diversity metrics that provide insight into the microbial diversity within a sample by considering both richness (the number of unique species) and evenness (the relative abundance of these species). In the samples analyzed, OTUs ranged from 108 to 148, indicating varying levels of microbial complexity across the samples. Sample AAA-111 S showed the highest OTU count (148), indicating a highly diverse microbial community, which is further supported by its elevated Shannon and Simpson index values, signifying not only high richness but also a more even distribution of species within the community.

Table 1—Sequencing statistics

SampleNam	DNA concentration	RawReads	FiltReads	MappedRea	MappedRati	SC	Richness	Shannon	Simpson	OTUs
AAA-111 S	0.616950721	30463	26520	10257	0.38676471	1	148	4.26538822	3.26104404	148
BBB-222 S	4.259530603	29183	25900	14582	0.56301158	1	108	3.50626065	2.36764973	108
CCCC-333 S	1.04367497	85002	77299	45035	0.5826078	1	122	3.65166625	2.42598035	122

Microbial taxonomy and classification

In all three soil samples, the class Actinobacteria was dominant, with relative abundances ranging from 48% to 73%. Sample BBB-222S exhibited the highest fraction of Actinobacteria (73%), followed by CCC-333S (67%) and AAA-111S (48%). This dominance of Actinobacteria across the samples suggests a strong adaptation of these bacteria to the oil-contaminated environments, which may contribute significantly to the degradation processes in the MECs. Firmicutes were present in all samples, with proportions ranging from 9% to 18%. The presence of Chloroflexi and Firmicutes in varying proportions across the samples may indicate their potential roles in diverse metabolic pathways essential for the contaminant's degradation. The soil sample AAA-111S displayed greater microbial diversity compared to BBB-222S and CCC-333S (Table 1, Figure 1a). This higher diversity in AAA-111S included notable abundances of Acidobacteria (3.4%), Gemmatimonadota. The presence of these microbial classes in oil-contaminated soils has been previously reported in the literature (Dastgheib et al., 2017), indicating their ability to adapt to harsh, hydrocarbon-rich environments. Their metabolic versatility enables them to degrade a wide range of contaminants, making them particularly valuable in bioremediation strategies. This microbial diversity could contribute to a more robust and stable MEC system, as a more varied community can maintain function across different environmental conditions and support complex metabolic interactions required for efficient contaminant removal.

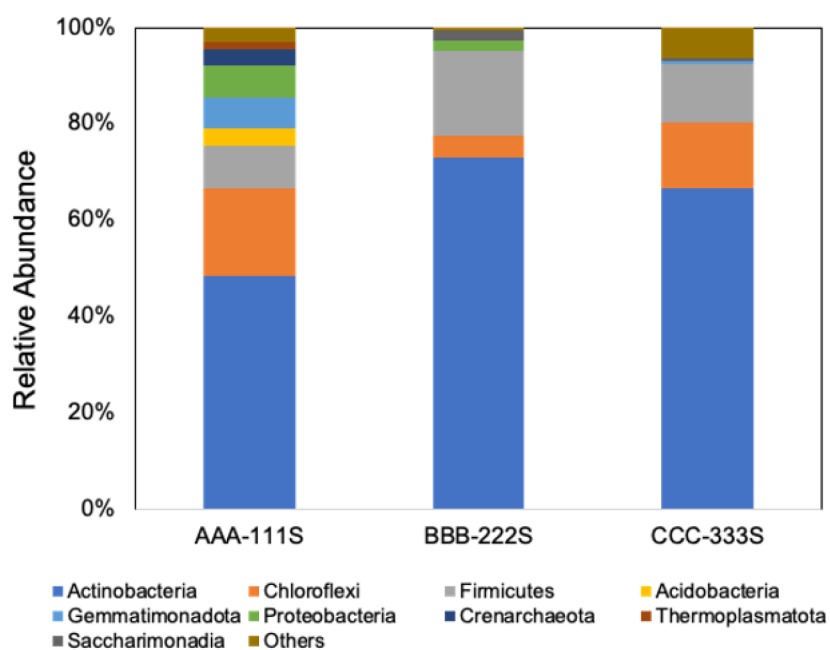


Figure 1a—Relative abundance of microbial class of the soil samples.

Setup and operation of MEC

The single-chamber MECs (Figure 3A) were set up in glass bottles with a working volume of 280 mL and a headspace of 20 mL. Gas bags (0.1 L Cali-5-Bond, Calibrate, Inc.) were connected to the top of the reactor through butyl rubber stoppers. The anode material was either carbon felt (2.5 cm × 6.5 cm) or carbon brush (2.5 cm diameter × 2 cm length; PANEX 33 fibers, ZOLTEK Inc., St. Louis, MO, USA), while the cathode was a titanium mesh (2.5 cm × 6.5 cm). The distance between the anode and cathode was 3 cm. The growth medium was supplemented with varying concentrations of PW and DW as the sole energy and inoculum source. The medium was boiled and cooled to room temperature while being sparged with an N₂:CO₂ (80:20, vol/vol) gas mix for 30 minutes to remove dissolved oxygen, followed by autoclaving. The MECs were operated at a voltage of 0.8 V using a power source (3645 A, Array, Inc.) in fed-batch mode at 30°C. When current generation dropped below 70% of its peak, the MECs were replenished with fresh feed, and a

new batch cycle was initiated. Electrical current in the circuit was determined by measuring voltage across a $10\ \Omega$ resistor at 15-minute intervals using data acquisition equipment (ADC-24; Pico Technology, UK). Two sets of MECs were tested: one set fed with increasing concentrations of PW (0.5-10%) and another set with increasing fractions of DW (0.5-10%) (Figure 3A). Control reactors, operated in a similar manner, had no applied voltage (open circuit) (Figure 3B). Additionally, PW and DW were inoculated in serum vials (Figure 3C) to enrich anaerobic microorganisms for potential use as inoculum in MEC experiments, tested at varying concentrations (10-100%). This strategy, involving different concentrations of PW and DW for inoculum enrichment and MEC operation in both open and closed circuits, aimed to identify optimal conditions for the enrichment of functional microorganisms capable of dual functions: hydrocarbon degradation and electrical current production. The experimental conditions also evaluated the role of microorganisms in petroleum hydrocarbon contaminant removal.

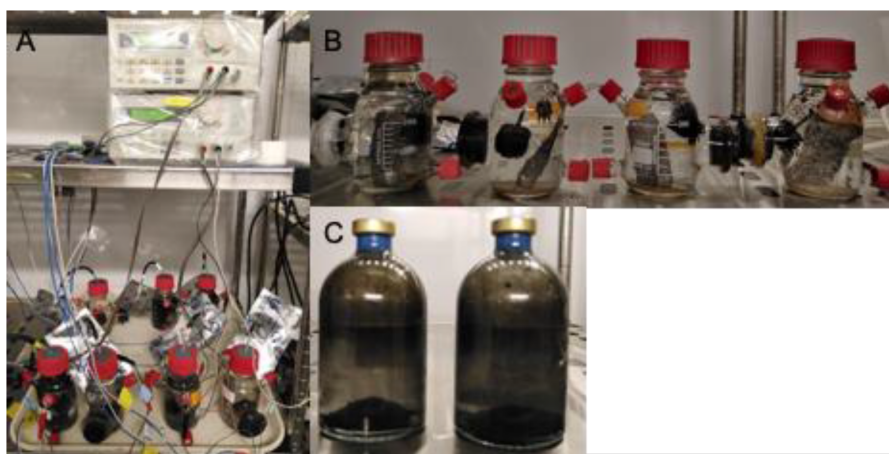


Figure 2—Single chamber MEC connected with power supply (A); MEC operated as open circuit (no voltage applied) (B); and PW and DW enrichment for anaerobic microorganisms in serum vials (C).

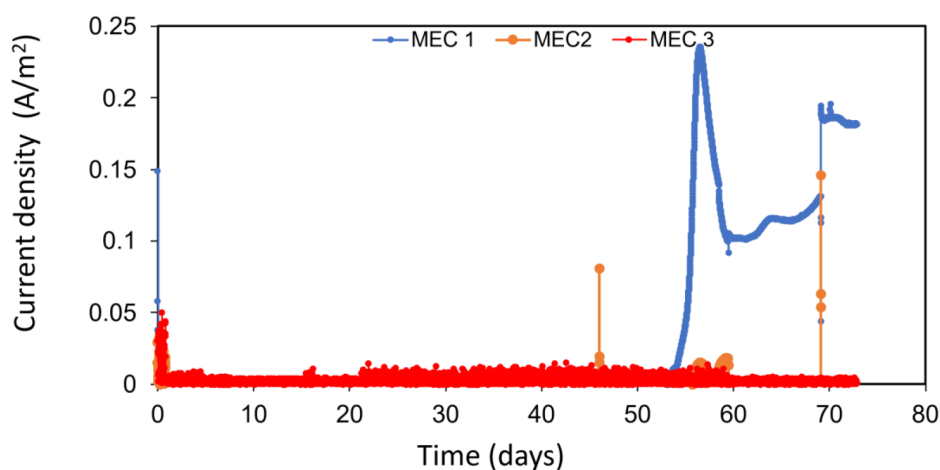


Figure 3—Current density of MEC. MEC-1 fed with DW and soil as inoculum; MEC-2 fed with DW and MEC-3 fed with PW.

Electrical current production of MEC reactors

During the first 45 days of MEC operation, none of the tested conditions (MEC-1, MEC-2, and MEC-3) exhibited electrical current production. At this stage, the concentrations of PW and DW were 0.5% and 2%, respectively, and graphite felt was used as the anode material. Due to the oily nature of PW and DW, the anode surface became coated with oil or viscous substances, likely hindering the attachment and proliferation of electroactive bacteria (EAB), which resulted in no electrical current generation. Additionally, the growth medium was prepared under oxygenic conditions, which may have further limited

EAB activity. Following this phase, several modifications were made. Carbon brush, which provides a higher surface area for EAB attachment, was selected as the new anode material. The growth medium was prepared under anaerobic conditions using an N₂:CO₂ gas mixture; and soil samples were added as an inoculum to the MEC-1 reactor. The impact of these changes is evident in the sharp increase in current density in MEC-1, which began around day 56. This rapid rise in current reached a peak of approximately 0.24 A/m², indicating successful EAB attachment and activity. The data suggest that the performance of MEC-1 was vastly improved by the adjustments made after day 45, while the current densities of MEC-2 and MEC-3 remained below 0.01 A/m², indicating that further modifications may be required to enhance their performance. These results indicate that the adjustments made after day 45 facilitated the attachment and growth of EAB, resulting in successful electrical current production. Further operation of the MEC reactors is expected to improve electrical current generation, hydrocarbon degradation, and coulombic efficiency.

Development and Optimization of Analytical Methods

Methods have been developed and optimized for measuring the concentrations of TPH, PAH, and BTEX in PW and DW. TPH concentrations are measured using GC-TCD-FID with a DB-1 column (100% dimethylpolysiloxane; 30 m × 0.25 mm I.D., 0.25 µm) and helium as the carrier gas at a flow rate of 1.5 mL/min. PAH concentrations are measured using Purge & Trap-GC-FID- NPD-MS with a DB-5 ms column (5% phenyl, 95% dimethylpolysiloxane; 30 m × 0.25 mm I.D., 0.25 µm) and helium at a flow rate of 1.5 mL/min. BTEX concentrations are measured using HT-GC-FID-MS with an Agilent J & W DB-624 UI column (30 m × 0.25 mm, 1.4 µm) and helium as the carrier gas at a flow rate of 10 mL/min. These established methods will be used to quantify the removal of TPH, PAH, and BTEX in MECs and serum vial reactors, facilitating the assessment of contaminant degradation in the treatment process.

Technology Value of MEC for PW treatment

Conventional biological wastewater treatment (WWT) systems are highly energy- intensive, typically consuming around 0.6-1 kWh/m³ of treated water. These systems do not recover valuable resources during the treatment process and result in significant operational costs. In contrast, MECs offer a much more energy-efficient solution, with energy consumption of approximately 0.05 kWh/m³. This reduction in energy use, combined with the potential for energy-positive treatment (i.e., generating more energy than is consumed), positions MECs as a highly attractive alternative for PW treatment in the oil and gas industry. The PW treatment estimates are presented in [Table 2](#).

Table 2—Treatment of PW with MEC estimates

Parameters	Scenario
COD removal	~70%
COD concentration	800–1400 mg/L
Anodic coulombic efficiency (fraction of the electrons in organics that ends up in electrical current)	~70%
Cathodic coulombic efficiency with Ni as catalyst for HER (percentage of electrical current converted to products at the cathode) Ni is a cheap cathode material for scale up	~60%
Hydrogen production	0.05–4.5 Kg of H ₂ produced/Kg of COD treated

The ability of MEC to operate in an energy-neutral or even energy-positive mode is a key differentiator compared to conventional WWT systems. By using electroactive bacteria to drive the conversion of organic matter into electrical current, MECs not only reduce energy consumption but also recover resources in the form of hydrogen gas (H_2). Hydrogen, a valuable byproduct, can be used as a clean energy source, further improving the energy balance of the process. The operational expenses (OPEX) for PW treatment are expected to be significantly reduced when using MEC technology (**Table 3**). While conventional systems incur high costs due to their energy demand and maintenance requirements, MECs have much lower energy consumption, which leads to direct cost savings. Based on the estimated energy consumption of ~ 0.05 kWh/m³, MECs could reduce energy-related operational expenses by up to 80-90%. Another advantage of the MEC technology is that the amount of residual sludge generated is significantly lower by 95% than the residual sludge generated from existing biological wastewater treatment technologies. Sludge management represents 50% of the total operational cost of wastewater treatment systems. For a large-scale operation, such savings could translate into millions of dollars, offering a highly cost-effective solution for PW treatment. In addition, MECs allow for a more streamlined treatment process, reducing the need for expensive chemicals or mechanical equipment required in traditional WWT systems. The reduced complexity and energy demand of MECs not only lower operational costs but also minimize the carbon footprint of PW treatment, contributing to Aramco's sustainability goals. Further cost reductions can be realized through the scalability of MEC systems, as modular reactors can be constructed with relatively low capital expenses (CAPEX), ranging from $\sim \$100$ – $\$300$ USD/m³ depending on materials used and the reactor configuration.

One of the most valuable aspects of MEC technology is its ability to generate hydrogen as a byproduct of the treatment process. Hydrogen production is directly linked to the chemical oxygen demand (COD) of the wastewater being treated. Based on typical PW treatment scenarios (**Table 2**) and 0.05–4.5 Kg H_2 produced/Kg COD treated, MECs can produce between 50–4500 kg of H_2 assuming a COD concentration of 1,000 mg/L and a treatment capacity of 1,000 m³/d of PW. This production capacity can generate significant revenue, especially when the price of hydrogen is estimated at 3–6 USD per kilogram. Optimization of the process parameters can further enhance MEC performance and production of hydrogen. Incorporating MECs into PW treatment operations would not only reduce operational costs but also create a new revenue stream. By recovering and selling hydrogen produced during the treatment process, the oil and gas industry could offset much of the treatment cost, potentially turning PW treatment from a cost center into a profitable operation. Furthermore, hydrogen produced in MECs has a purity of around 98–100%, which positions it as a high-value product compared to hydrogen generated by other technologies, which often require additional purification steps. The long-term durability of MECs also adds to their value proposition. With an expected lifespan of over 10 years, MEC pilot plants are designed to be robust and low-maintenance. The simplicity of the design, coupled with the use of durable materials, ensures that MEC systems can operate for extended periods with minimal intervention. This longevity further reduces OPEX and ensures a strong return on investment (ROI), particularly when compared to conventional systems that may require more frequent maintenance and equipment replacement. The potential for energy-neutral or energy-positive operation, combined with substantial cost savings, hydrogen production, and long-term durability, makes MECs a valuable addition to the wastewater treatment infrastructure.

Conclusion

The results of this study demonstrated several key findings: (i) carbon brush was identified as a suitable anode material for producing electrical current in MECs fed with 447 PW and DW, highlighting its high surface area and durability, which are crucial for enhancing the colonization of electroactive bacteria and facilitating efficient electron transfer, (ii) degassed growth medium effectively supported electrical current production in MECs fed with PW and DW, ensuring an optimal anaerobic environment that

promotes microbial activity and limits the presence of oxygen, a known inhibitor of electroactive biofilm development, (iii) the addition of soil as an inoculum resulted in a current density of 0.24 A/m² in MECs fed with DW, source of diverse microorganisms with functional capabilities for hydrocarbon degradation and electron transfer, (iv) varying concentrations of PW and DW (0.5-3%) generated electrical current in MECs, indicating their potential as energy sources, demonstrating that both PW and DW contain sufficient organic matter to support bioelectrochemical processes, which could be further optimized for larger-scale applications, (v) soil samples contained diverse microbial communities capable of degrading a wide range of petroleum hydrocarbons, including members of Actinobacteria, Chloroflexi and Firmicutes, suggesting microbial and metabolic versatility of these microbial communities, (vi) methods for quantifying the removal rates of TPH, PAH, and BTEX in MEC and serum vial reactors were developed and optimized, ensuring accurate tracking of contaminant breakdown and offering a reliable approach to assess the efficiency of the treatment process, and (vii) the economic viability of MEC technology for PW treatment, including projections for revenue, CAPEX, and OPEX, was assessed, laying the groundwork for evaluating the cost-effectiveness and scalability of MECs as a sustainable solution for treating PW while recovering energy in the form of hydrogen.

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References

1. Cabrera, J., Irfan, M., Dai, Y., Zhang, P., Zong, Y., & Liu, X. (2021). Bioelectrochemical system as an innovative technology for treatment of produced water from oil and gas industry: A review. *Chemosphere*, **285**, 131428. <https://doi.org/10.1016/j.chemosphere.2021.131428>
2. Ghasemi Naraghi, Z., Yaghmaei, S., Mardanpour, M.M., & Hasany, M. (2015). Produced Water Treatment with Simultaneous Bioenergy Production Using Novel Bioelectrochemical Systems. *Electrochimica Acta*, **180**, 535–544. <https://doi.org/10.1016/j.electacta.2015.08.136>
3. Olajire, A.A. (2020). Recent advances on the treatment technology of oil and gas produced water for sustainable energy industry-mechanistic aspects and process chemistry perspectives. *Chemical Engineering Journal Advances*, **4**, 100049. <https://doi.org/10.1016/j.cej.2020.100049>
4. Al-Taqi, A.A., Al-Dahlan, M.N., & Alrustum, A.A. (2017). Maintaining Injectivity of Disposal Wells: From Water Quality to Formation Permeability. D031S025R004. <https://doi.org/10.2118/183743-MS>
5. Katuri, K.P., Ali, M., & Saikaly, P.E. (2019). The role of microbial electrolysis cell in urban wastewater treatment: Integration options, challenges, and prospects. *Current Opinion in Biotechnology*, **57**(120 L), 101–110. <https://doi.org/10.1016/j.copbio.2019.03.007>
6. Mohanakrishna, G., Al-Raoush, R.I., Abu-Reesh, I.M., & Aljaml, K. (2019). Removal of Petroleum hydro-carbons and sulfates from produced water using different bioelectrochemical reactor configurations. *Science of The Total Environment*, **665**, 820–827. <https://doi.org/10.1016/j.sci-totenv.2019.02.181>
7. Mohanakrishna, G., Al-Raoush, R.I., & Abu-Reesh, I.M. (2021). Integrating electrochemical and bioelectrochemical systems for energetically sustainable treatment of produced water. *Fuel*, **285**, 119104. <https://doi.org/10.1016/j.fuel.2020.119104>
8. Bader, M.A., Hari, A.R., Katuri, K.P., & Saikaly, P.E. (2023). Effect of start-up strategies with functionally redundant and efficient acetoclastic electroactive bacteria on the performance of

- microbial electrol-ysis cells fed with domestic wastewater. *BioresourceTechnology Reports*, **21**, 101323. <https://doi.org/10.1016/j.biteb.2022.101323>
9. El Kik, O., Issa, L., Katuri, K.P., Saikaly, P.E., Alameddine, I., & El-Fadel, M. (2021). Coupling anaerobic fluidized membrane bioreactors with microbial electrolysis cells towards improved wastewater re-use and energy recovery. *Journal of Environmental Chemical Engineering*, **9**(5), 105974. <https://doi.org/10.1016/j.jece.2021.105974>
 10. Shehab, N.A., Ortiz-Medina, J.F., Katuri, K.P., Hari, A.R., Amy, G., Logan, B.E., & Saikaly, P.E. (2017). Enrichment of extremophilic exoelectrogens in microbial electrolysis cells using Red Sea brine pools as inocula. *Bioresource Technology* <https://doi.org/10.1016/j.biortech.2017.04.122>
 11. Katuri, K.P., Kalathil, S., Ragab, A., Bian, B., Alqahtani, M.F., Pant, D., & Saikaly, P.E. (2018). Dual-Function Electrocatalytic and Macroporous Hollow-Fiber Cathode for Converting Waste Streams to Valuable Resources Using Microbial Electrochemical Systems. *Advanced Materials*, **30**(26), 1707072. <https://doi.org/10.1002/adma.201707072>
 12. Rousseau, R., Dominguez-Benetton, X., Délia, M.-L., & Bergel, A. (2013). Microbial bioanodes with high salinity tolerance for microbial fuel cells and microbial electrolysis cells. *Electrochemistry Communications*, **33**, 1–4. <https://doi.org/10.1016/j.elecom.2013.04.002>
 13. Kondaveeti, S., Govindarajan, D., Mohanakrishna, G., Thatikayala, D., Abu-Reesh, I.M., Min, B., Nambi, I.M., Al-Raoush, R.I., & Aminabhavi, T.M. (2023). Sustainable bioelectrochemical systems for bio-energy generation via waste treatment from petroleum industries. *Fuel*, **331**, 125632. <https://doi.org/10.1016/j.fuel.2022.125632>
 14. Krishna P. Katuri, Muhammad Ali, Pascal E. Saikaly. The role of microbial electrolysis cell in urban wastewater treatment: integration options, challenges, and prospects, *Current Opinion in Biotechnology*, 2019, ISSN 0958-1669, <https://doi.org/10.1016/j.copbio.2019.03.007>
 15. Produced Water Treatment Market Size And Forecast. (2022). Verified Market Research. <https://www.verifiedmarketresearch.com/product/global-produced-water-treatment-market/>
 16. Robeson, M.S., O'Rourke, D.R., Kaehler, B.D., Ziemski, M., Dillon, M.R., Foster, J.T. & Bokulich, N.A. (2020) RESCRIPT: Reproducible sequence taxonomy reference database management for the masses. bioRxiv, 2020.10.05.326504.
 17. Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., Peplies, J. & Glöckner, F.O. (2013) The SILVA ribosomal RNA gene database project: Improved data processing and web-based tools. *Nucleic Acids Research* **41**, 590–596. Doi:[10.1093/nar/gks1219](https://doi.org/10.1093/nar/gks1219)
 18. Yilmaz, P., Parfrey, L.W., Yarza, P., Gerken, J., Pruesse, E., Quast, C., Schweer, T., Peplies, J., Ludwig, W. & Glöckner F.O. (2014) The SILVA and "all-species living tree project (LTP)" taxonomic frameworks. *Nucleic Acids Research* **42**, D643–D648. <https://doi.org/10.1093/nar/gkt1209>
 19. Li H. (2018) Minimap2: Pairwise alignment for nucleotide sequences. *Bioinformatics* **34**, 3094–3100. DOI: [10.1093/bioinformatics/bty191](https://doi.org/10.1093/bioinformatics/bty191)
 20. Danecek, P., Bonfield, J.K., Liddle, J., Marshall, J., Ohan, V., Pollard, M.O., Whitwham, A., Keane, T., McCarthy, S.A., Davies, R.M. & Li, H. (2021) Twelve years of SAMtools and BCFtools. *GigaScience* **10**, 1–4. DOI: [10.1093/gigascience/giab008](https://doi.org/10.1093/gigascience/giab008)
 21. Albertsen, M., Karst, S.M., Ziegler, A.S., Kirkegaard, R.H. & Nielsen, P.H. (2015) Back to basics-the influence of DNA extraction and primer choice on phylogenetic analysis of activated sludge communities. *PLOS ONE* **10**, e0132783. <https://doi.org/10.1371/journal.pone.0132783>
 22. Hsieh, T.C., Ma, K.H. & Chao, A. (2016) iNEXT: An r package for rarefaction and extrapolation of species diversity (hill numbers). *Methods in Ecology and Evolution* **7**, 1451–1456. DOI: <https://doi.org/10.1111/2041-210X.12613>

23. Chao, A., Gotelli, N.J., Hsieh, T.C., Sander, E.L., Ma, K.H., Colwell R.K. & Ellison A.M. (2014) Rarefaction and extrapolation with hill numbers: A framework for sampling and estimation in species diversity studies. *Ecological Monographs* **84**, 45–67. <https://doi.org/10.1890/13-0133.1>